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Chapter 24: The Evolution of Financial Knowledge Production—Judgment, Models, and Market Feedback

Abstract

Knowledge in financial markets is not a single uniform product. Analysts form judgments about future returns; researchers write part of those judgments into models and computational procedures; trading generates prices; and institutions then use these prices to evaluate both investors and knowledge tools. Taking modern financial research since the mid-twentieth century and its adoption in the U.S. market as its main thread, this chapter examines how judgment, calculation, readings, and certification became connected. Markowitz separated security analysis from portfolio computation and preserved entry points for testing approximations and sample selection; the work of securities price databases and sell-side analysts shows that computable inputs are themselves products of institutional production. Risk adjustment, index products, paper pricing sheets, implied volatility, and automated quotation systems demonstrate how tools acquired the respective qualifications of comparison, investment, coordination, and professional recognition. The distinctive difficulty of financial knowledge is not merely that judgment is compressed into models and readings, but that the use of tools participates in forming the very market evidence later used to evaluate them: spread trading may improve model fit; publication may shrink original return opportunities; shared blind spots may replace independent verification with mutual confirmation; and collective adoption may alter the execution conditions of protective strategies. Adoption, market fit, professional recognition, and knowledge of risk must therefore be separately evidenced. The continuous case of

portfolio insurance further shows that local protection, collective consequences, client outcomes, and post-crisis corrections cannot be merged into a single label of success. Knowledge growth takes the form of new materials, effective counterevidence, clarified boundaries of applicability, and the correction of errors along chains of use; organizational coordination, commercial success, and improved execution conditions can support these outcomes, but cannot automatically prove them. This historical analysis provides comparative questions for the knowledge production of platforms and artificial intelligence (AI), without presuming that different domains share the same failure mechanisms.

Keywords

financial knowledge production; judgment and calculation; computable inputs; knowledge certification; market evidence; fiber; mean-variance analysis; market feedback; portfolio insurance; execution conditions

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§1 The Object of Study: How Financial Knowledge Acquires the Qualification for Action

§1.1 Judgment, Calculation, and Transaction Records Are Not the Same Kind of Knowledge

An investment report, a pricing equation, and an executed transaction price can all enter financial decisions, yet they do not share the same evidentiary identity. An investment report makes judgments about firms, policies, and changes in demand; a pricing equation specifies how to compute under given assumptions; a transaction price records a deal reached at a specific moment under specific conditions. The three can enter into one another, but they cannot be arranged into a single-line history in which theory grows ever more precise and the market thereby comes ever closer to knowing reality.

The financial knowledge production discussed in this chapter comprises three connected activities. Analysts, investors, and research teams organize observations and experience into judgments about the future; academic researchers and financial engineers write part of those judgments into models, parameters, and computational procedures; market participants use these tools to organize trades, and the outcomes become new public readings that enter the next round of research, investment, and evaluation. “Knowledge” here is a name for the object of study; it does not imply that every adopted judgment has thereby been shown to be true.

Expectation, in this chapter, refers to beliefs about what has not yet happened, including judgments about returns, risks, and conditions for action. It is not identical with the technical definition of rational expectations in any particular school, and it does not merge the belief and the position built on it into one object. The same belief can yield different positions depending on funding, liabilities, and trading constraints; the same position can also be supported by different beliefs. Markowitz’s division of labor between probability beliefs and portfolio choice provides the starting point for studying this relation (Markowitz, 1952, 1959, 2002, 2020).

Markets are not devices that produce nothing but prices. Research reports, risk classifications, databases, contracts, computational methods, and professional services jointly determine how prices can enter action. Part of the information contained in prices is the result of these activities; one cannot first treat prices as a final judge independent of knowledge production, and then use them to certify unconditionally that every link of that production is correct.

§1.2 The Central Question of This Chapter: How Use Changes Subsequent Evidence

This chapter traces a process: judgments enter models, models enter investment and trading, market outcomes enter the evaluation of tools, and evaluation in turn shapes subsequent adoption. Each link has its own producers, materials, and constraints. What must be answered is not which theory ultimately replaced the others, but how knowledge tools were proposed, adopted, certified, and revised.

"Certification" is used here for a trackable action: researchers, clients, management, or regulators, on the basis of some performance, grant a tool or its users a corresponding degree of credibility and eligibility for use. It is not restricted to formal credentials. Empirical testing, client mandates, and regulatory approval are three different procedures; a theory being discussed, a service being procured, and positions being adjusted are also three different degrees of adoption.

Whether certification overreaches depends on what the user claims the tool can accomplish. If the tool promises only to reduce the cost of a certain computation, its results should not be negated on the ground that it has not eliminated all risk; evidential overreach occurs when evidence is used to support a capability claim beyond the scope of the test—for example, when price fit within a finite sample is used to warrant execution capability that has not been tested. A history of adoption can show that a method was used, but not that all users drew the latter inference.

The distinctive difficulty of financial knowledge is therefore not merely that judgment is compressed into models and readings, but that the use of tools participates in forming the very market evidence later used to evaluate them. Prices drawing closer to models, original return opportunities shrinking, others' positions providing reassurance, and collective execution conditions changing are mechanisms running in different directions; they cannot be merged into a single cyclical fable of "model success causing failure."

§1.3 Fiber Analysis: What Other States Remain Compatible with the Same Reading

The Clock Room of Chapter 1 distinguishes readings from the states that the readings are meant to characterize. Financial analysis does not face a fully closed room: business operations, cash flows, contract performance, trade execution, and actual losses can all impose constraints; the question is through what interpretations these constraints pass, and whether they can correct existing judgments.

The Fiber Framework of Chapter 4 formalizes this inquiry as an analysis of compatible states. For a given mapping function (π) and reading (r), the fiber is the set of all true states that can produce that reading:

$$\pi^{-1}(r) = \{T' \mid \pi(T') = r\}.$$

Here the candidate true states (T') can include business operations, return paths, financing conditions, participants' positions, and other factors relevant to the question at hand. Where the mapping is partly unknown, this chapter discusses compatible states under the knowledge constraints already stated, without describing an observer's added knowledge as a change in the fixed mapping itself. The work here is qualitative discrimination, not a claim to compute the volume of the entire financial system's fiber.

Concrete analysis must also fix the task, the observer, and the point in time. Evaluating whether a computation is correct, identifying how a protection is fulfilled, and forecasting future losses do not face the same set of

states. Historical outcomes can help trace mechanisms of failure, but they cannot be retrospectively attributed to the participants as knowledge they possessed at the time.

For example, the same investment return can be compatible with correct judgment, with more risk taken on, or with a fortuitously favorable price move. For the client evaluating investment skill, the return is a visible reading; the mechanism that produced it need not be equally visible. If the latter two explanations have not been ruled out, yet the return is treated as sufficient proof of skill, then the risk of the pseudo-small fiber appears: visible success creates the false impression that few relevant possibilities remain unresolved. Here the observer must be specified as the client; the client's ignorance does not license the conclusion that the managers and all researchers are equally ignorant.

The fiber and the pseudo-small fiber must also be kept apart. Different mechanisms producing the same return is an identification problem; over-interpreting such returns as skill is another problem. An institution may know full well that return attribution is insufficient and still keep using a metric because of client assessment. What should then be studied is assessment and resource allocation; one cannot infer, merely from compliance with a metric, that participants believe the metric suffices to explain all the risk.

§1.4 From Compatible States to Historical Process

Statically listing several explanations is not enough. Chapter 17 distinguishes the functional recursion in which prices guide allocation from the pathological recursion in which prices reference themselves, and tracks the slippage between the two. This chapter applies that distinction to the adoption of knowledge tools: does feedback keep checking the tool against its task, or does it let certain local successes expand usage while excluding shared execution conditions and the costs of failure from evaluation? The latter is a risk mechanism that calls for investigation, not a universal law that every use of a reading must produce.

Chapter 23 has already distinguished publication, recognition, and knowledge growth, and has shown that knowledge growth can take the form of new materials, effective counterevidence, clarified boundaries of applicability, and the correction of errors. This chapter takes up the division of labor among judgment, calculation, and readings, and studies a more direct feedback: the use of knowledge tools may participate in changing the very markets they subsequently face. Chapter 25 continues to study how organizational relations shape purposes, beliefs, habits, and conditions for action; this chapter stops at knowledge evidence and the authority to judge, and does not merge family authorization, identity formation, and investment life history into a single concept of validity.

Chapter 30 currently organizes financial history under the heading "judgment replaced by models." That formulation can only serve as a proposition to be examined, not as a prescription for this chapter's conclusions. This chapter will distinguish among judgment being redistributed, certain computational results acquiring substitute qualifications, and whether the capacity for judgment has disappeared as a whole. These are not the same proposition.

§1.5 Five Kinds of Success and Their Evidentiary Boundaries

Pricing sheets, risk reports, and shared quotation systems enable financial knowledge to acquire different qualifications for use. The five kinds of success listed in Table 1 are not a single scale of increasing completeness, but judgments that must be separately evidenced. What fiber analysis asks is: for the task at hand, which states

has this achievement actually ruled out, and which states remain compatible.

Table 1. Five Kinds of Success for Financial Knowledge Tools and Their Evidentiary Boundaries

Meaning of success	Evidence already obtained	What cannot be inferred from this alone
Computable	Given inputs and constraints, results can be obtained	Inputs suit the future; the problem has been fully expressed
Tradable	Positions can be established under the relevant rules, financing, and execution conditions	Exit at the same cost remains possible under stress
Profitable	Positive returns achieved in a given period under a given accounting convention	Returns stem entirely from knowledge advantage; tail risk has vanished
Fitting market prices	The model shows small deviation under specified samples and parameters	Parameters have independent predictive power; fit is unaffected by model use
Identifying specific risks	Specific risks have been identified with justification and tested accordingly	Other risks are covered; future institutional and market conditions will not change

Computability, tradability, and profitability are themselves important achievements. If they are used to support capability claims beyond the corresponding tasks, the formation of the evidence must be checked further. A computational program having been correctly executed does not simultaneously prove that orders can be filled under stress; if the mandate explicitly requires only the former, the latter's non-completion cannot be written as the program's failure.

Returns can constrain error, but they are not a one-time verdict on all judgments. A strategy whose risks were fully disclosed may lose money; a strategy that concealed tail risk may profit continuously. Only by linking prior claims, positions, risk exposures, costs, and actual outcomes can one judge what success or failure means for a knowledge claim.

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§2 How Computable Inputs Are Produced

§2.1 Portfolio Selection First Drew a Boundary

Markowitz's 1952 portfolio selection study changed one analytical unit: investment risk cannot be assessed by looking only at how each security in a portfolio fluctuates individually; one must also look at how they vary together. Holding many securities does not automatically constitute effective diversification; portfolio risk depends on weights, individual risks, and their interrelations. This change enabled analysts to compare entire portfolios rather than merely rank a set of favored securities (Markowitz, 1952, 1959).

This work did not simultaneously solve the problem of how future returns are to be known. In his 2002 retrospective, Markowitz circumscribed his own task very clearly:

> I assumed that it was the job of the security analyst to provide the inputs to the portfolio analysis. I took it as my task to figure out how to derive efficient sets from these inputs. (Markowitz, 2002)

This division of labor does not declare judgment to be irrelevant. It makes a conditional computation possible: once inputs and constraints are given, one can compare which portfolios are dominated by others in terms of risk and expected return, and which lie on the efficient frontier. Whether the inputs are well-grounded must still be answered by security analysis and subsequent testing.

This chapter accordingly distinguishes two kinds of knowledge product. Analysts' estimates of future co-movements can be good or bad; computational procedures can be correct or erroneous. Executing a procedure accurately does not automatically improve the reliability of input estimates; insufficient inputs do not mean the procedure is mathematically wrong. Conflating this division of labor makes it easy to blame all investment losses on the equation, or conversely to use the equation's correctness as a guarantee for all investment judgments.

§2.2 Historical Data Is Research Infrastructure Built by Institutions

"Analysts provide the inputs" does not mean that market history already exists in computable form. When Fisher and Lorie published their study of securities returns in 1964, they began by describing the facilities and plans of the Center for Research in Security Prices, rather than merely reporting a set of return rates. Their research shows that systematic comparison depends on a research environment composed of data, computer tapes, and programs (Fisher & Lorie, 1964).

This infrastructure embodies concrete production work. The Center organized monthly closing prices of common stocks on the New York Stock Exchange onto magnetic tapes and checked them repeatedly; capital-change records included cash dividends, stock dividends, stock splits, rights issues, and share exchanges. For price series to be usable in fine-grained analysis, these changes had to be encoded, verified, and their dates and tax attributes preserved. Data did not flow directly from the market into equations; it had to be collected, classified, corrected, and stored before it could become computable input.

This distinction changes how historical evidence is evaluated. Commercial funding can support data production, but it cannot by itself prove that data has been contaminated; the sheer scale of data cannot substitute for checking the encoding and verification procedures. What must genuinely be asked is: which records were preserved, how capital changes were handled, how errors were detected, which materials have not yet been integrated, and whether researchers can re-examine these steps. The database is thus both a knowledge product and an evidentiary condition for subsequent computation.

This boundary had been articulated much earlier in economic thought. In *Risk, Uncertainty and Profit*, Knight distinguished measurable uncertainty from unmeasurable uncertainty: the former covers situations in which the distribution of outcomes can be known—for these Knight reserved the term "risk," transferable through grouping and insurance; the latter covers situations in which that distribution cannot be known, because the situation is too unique to be grouped. He wrote:

> The practical difference between the two categories, risk and uncertainty, is that in the former the distribution of the outcome in a group of instances is known (either through calculation a priori or from statistics of past experience), while in the case of uncertainty this is not true, the reason being

> in general that it is impossible to form a group of instances, because the situation dealt with is in a high degree unique. (Knight, 1921)

By this distinction, grouping is the precondition of computability: only situations that can be assimilated into classes of instances, with the proportions of their outcomes computed, can become statistical inputs. The monthly closing prices and capital-change records that the Center for Research in Security Prices repeatedly checked and encoded dealt precisely with groupable situations; the situations Knight described as ungroupable would not, conceptually, enter such a database at all. The boundary of the database is thus not merely a matter of engineering—not only which materials have yet to be integrated—but conceptual and older: what cannot be grouped cannot enter computation as probability (Knight, 1921).

§2.3 Analysts Produce More Than Forecast Numbers

Professional judgment is likewise not a transparent raw material waiting to be loaded into models. Brown et al. (2015) surveyed 365 sell-side analysts and conducted 18 follow-up interviews, studying earnings forecasts, stock recommendations, industry knowledge, management communication, compensation, and career evaluation. Their research shows that forecasts are often a means to stock recommendations, not the sole endpoint of professional work (Brown et al., 2015).

Analysts' work also includes industry knowledge, communication with company management, and helping clients understand firms. The study found that private communication with management was an important input to forecasts and recommendations, and that industry knowledge was highly valued by buy-side clients. This means that forecast error covers only part of professional output: it can check whether a particular estimate was accurate, but it cannot by itself evaluate whether industry explanation, question clarification, and client decision support were effective.

Professional recognition and knowledge checking are also not the same thing. The study simultaneously found that earnings forecasts and stock recommendations significantly below consensus could often enhance rather than undermine analysts' credibility among investment clients. Analysts do not merely follow consensus; client recognition sometimes rewards demonstrably independent judgment. But client approval still cannot substitute for substantive verification of forecast bases, communication materials, and actual recommendations.

This body of material reconnects input production to professional institutions: who can access management, which industry knowledge is preserved, which judgments are written into forecasts, and which outputs gain client and employer recognition—all of these shape what enters the model. Analysts provide more than numbers, yet models often receive only numbers. Compression can be the legitimate cost of division of labor, but the numbers themselves no longer carry the full conditions of production.

§2.4 Approximation Error and Prediction Error Must Be Separated

Mean-variance analysis organizes portfolio selection around the mean, variance, and covariance of returns. An important defense of it is that, under certain conditions, these quantities can reasonably approximate an investor's utility evaluation of a return distribution. This involves both computational gain and informational trade-off, and both sides must be counted (Markowitz, 1991).

Using return (R), utility function (U), expected return (E), and return variance (V), under the relevant smoothness and moment conditions, the second-order approximation function (f) of expected utility can be written

as:

$$\mathbb{E}[U(R)] \approx f(E, V) = U(E) + \frac{1}{2}U''(E)V.$$

In the formula, expected return is the mean of the return distribution, return variance describes dispersion relative to that mean, and the second derivative of the utility function reflects local curvature. Whether the approximation is good enough depends on the utility function, the return distribution, and the impact of the neglected parts on decisions; the formula's simplicity is not evidence of the approximation's validity.

In his Nobel lecture, Markowitz discussed computational comparisons of the historical return distributions of 149 mutual funds. Under the various utility functions examined, the correlation between approximation values and expected utility was high, but there were also cases where the approximation was poor. He also discussed how to check the deviation between a utility function and its quadratic approximation (Markowitz, 1991). These results provide checkable grounds for certain approximations; they are not a universal guarantee for all investors, all future markets, and all return scales.

Approximation error and prediction error must also be distinguished. Even if an approximation is excellent on a given distribution, the analyst may still have chosen the wrong distribution to describe the future. A computational method can faithfully process a set of inapplicable inputs; enlarging the historical sample cannot by itself prove that the conditions covered by the sample will persist into the next phase. What the tool can do, and how to check what it has not done, must be preserved in the same account.

§2.5 Responsibility for Application Cannot Be Canceled by the Label "Framework"

Mean-variance analysis provides a way of organizing input estimation, constraints, and portfolio comparison, not a law that unconditionally predicts future prices. Whether the derivations within a framework are valid and whether the framework suits a particular decision are different questions. If key liabilities, liquidity needs, or tail consequences have not entered the analysis, the computation may still be correct, but whether it is useful enough for the investor in question requires fresh judgment.

The handoff between judgment and calculation is therefore not a one-time event. A team writes judgments into parameters, the computation reveals certain allocation consequences, the team adjusts constraints accordingly, and then observes actual execution. This cycle can improve decision quality. When the sources of parameters and constraints are hidden and the efficient frontier is treated as a final answer requiring no further judgment, the conditionality of the tool is lost in transmission.

An analytical contrast can illustrate why the task must be fixed first. Imagine two funds holding identical assets and facing the same price path: one has no large interim payments, while the other must make payouts during a downturn. Reporting only the annual return of the assets would merge two different performance situations. The latter might need to sell assets, obtain financing, or adjust other allocations; its problem is not an inability to compute annual returns, but that this reading alone does not say whether the required funds will be available on payment dates. This contrast is not a historical case, but an illustration of why returns, paths, and obligations must all be fixed before evaluating a tool.

Adding payment schedules and liquidity needs to the analysis can increase discriminatory power; they are not in principle unquantifiable. However, deciding which payments must be prioritized, how large a shortfall is

acceptable, and who bears the consequences cannot be derived from the historical mean and variance of the assets alone. Reducing tracking error can be a management task; paying on time is another task; completing the former does not mean the latter has also been completed.

§2.6 Historical Data Has Not Made Judgment Disappear

Using historical returns is an important path for forming estimates. But the choice of which historical period, which assets, which frequency, and how to handle anomalous periods all affect the estimates. Markowitz states this explicitly:

> Even if one wanted to avoid any subjectivity in one's portfolio analysis by basing estimates strictly on historical returns without any subjective input, the choice of historical period greatly alters estimates and is thus, itself, a subjective forward-looking decision. (Markowitz, 2020)

Subjectivity here does not mean arbitrariness. A team can explain its sampling rationale, compare alternative windows, check the sensitivity of conclusions to input changes, and allow actual results to overturn prior judgments. What deserves criticism is not the existence of choice, but choice being packaged as though there were no choice, making it difficult for alternative data or alternative specifications to qualify for discussion.

The same set of parameters can correspond to different data sources and selection processes. An investor who receives only the final weights sees a completed computational result; what may be invisible is which periods the parameters are especially sensitive to. Disclosing more decimal places cannot substitute for these explanations; disclosing the basis of estimation, alternative scenarios, and limiting conditions is what enables the recipient to discern the differences originally merged into the same reading.

§2.7 Inputs Enter Computation with Their Conditions of Production

Data infrastructure, professional judgment, conditional computation, and application constraints form a continuous process. Historical returns are first encoded and verified to become research data; analysts then organize industry knowledge, management communication, and public records into forecasts and recommendations; computational procedures subsequently compress part of this into parameters; and the applicator must further check liabilities, liquidity, and actual execution.

This chain does not require every step to preserve all information, but it does require the recipient to know which information has been lost. If one sees only the final weights, one cannot judge from the weights alone whether the data is reliable, whether the forecasts are independent, whether the approximation is applicable, and whether payment obligations have been included. The next section answers how these computations, carrying their conditions of production, further acquire the qualifications of comparison, investment, and professional recognition.

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§3 How Tools Acquire Qualifications for Use and Professional Recognition

§3.1 Risk Adjustment Becomes a Service for Evaluating Investment Managers

MacKenzie (2006) records that an important entry point for portfolio and pricing theory into the industry was performance evaluation. Treynor's analysis pointed out that certain outstanding results could be explained by

portfolios with high market-risk sensitivity: achieving higher returns in rising markets also means potentially greater losses in falling markets. Risk adjustment thus added a comparison that raw return readings alone did not provide, enabling investors to re-examine managers' claims of skill.

This work subsequently formed professional services. John O'Brien studied Sharpe's methods and designed systems for evaluating investment managers; Becker Securities and other institutions also began offering related services. When databases and computing resources were not yet widespread, data on securities' market-risk sensitivity could be compiled into booklets for sale, giving institutional clients comparative readings they could not easily produce independently. MacKenzie also records the resistance of financial practitioners to these evaluations and the increased institutional interest in risk measurement after market declines (MacKenzie, 2006).

The change here has a dual character. On one hand, the new evaluation tools enabled clients to test skill claims based solely on raw returns. On the other hand, the standards for measuring skill gradually entered managers' business competition: managers had not only to achieve returns, but to explain those returns in the language of risk and benchmarks accepted by clients. Knowledge tools began to serve simultaneously the functions of research, oversight, and professional credential display.

A theory being discussed, a service being procured, and positions being adjusted are three different degrees of adoption; reports arriving at desks and meetings does not mean decisions have actually changed. Risk-adjusted returns also still depend on the chosen risk model and comparison benchmark; one cannot declare that skill identification has been completed on this basis.

Knight's distinction between measurable and unmeasurable uncertainty—situations that can be grouped, whose outcome distributions can be computed, belong to the former; situations too unique to be grouped belong to the latter—gives a conceptual name to this limit (Knight, 1921). Risk adjustment deals with uncertainty whose distribution can be known; ungroupable, unmeasurable uncertainty lies outside its coverage. This is a limit different from dependence on the chosen risk model and comparison benchmark: the former concerns how comparisons proceed within a chosen model, the latter concerns the fact that some objects never enter the comparison at all. A manager's high risk-adjusted return shows achievement with respect to measurable risk; whether that achievement extends to judgment about unmeasurable uncertainty remains an open question.

§3.2 The Evaluation Benchmark Itself Carries Model Conditions

The Capital Asset Pricing Model (CAPM) advanced portfolio analysis to market equilibrium. Sharpe (1964) adopted assumptions such as common borrowing conditions and homogeneous expectations in his derivation, so that investors evaluated investment opportunities using the same expected returns, standard deviations, and correlation coefficients. This made it possible to discuss how an asset's expected return relates to its contribution to the market's common risk.

Homogeneous expectations temporarily exclude differences among investors' views of opportunities from the derivation. This limits the phenomena the model can explain, but it does not prove that judgment has been deleted in reality. What the model assumes about how investors see things is not the same as what institutions actually require everyone to see; the latter change can only be explained by examining adoption and assessment institutions.

Sharpe explicitly acknowledged that these assumptions were highly restrictive and defended his research method by the acceptability of its implications. This differs in emphasis from Markowitz's work on checking approx-

imation errors, but it is not direct evidence that one side preserved judgment while the other eliminated it in history. Both must face the same follow-up question: once the comparative relations offered by theory entered institutions, what were they actually used to answer?

Empirical testing also requires specifying what counts as excess return. Using CAPM for risk adjustment and then judging whether market inefficiency exists tests not only market efficiency but also the chosen risk model. The market portfolio itself needs a proxy; a stock index is not the same as all risky assets. MacKenzie's (2006) review of this debate shows that models, data, and auxiliary assumptions all enter the test, and that the attribution of anomalous results can therefore be disputed.

The difficulty of joint testing requires making clear what has been rejected and what alternative explanations remain. If changing the benchmark, period, or risk adjustment changes the conclusion, that sensitivity itself is something users need to know; one cannot keep only the single test most favorable to the tool.

§3.3 The Index Turns from Comparison Benchmark into Investment Product

Performance evaluation also drove another transformation: if active management could not consistently beat the index after costs, could the index, originally a comparison object, itself become an investment strategy? Index products required the simultaneous presence of investment institutions, client demand, personal connections, and execution capability.

John A. McQuown of Wells Fargo Bank brought financial economics researchers into the bank's advisory and research activities. MacKenzie records that, after Sharpe made the connection, Samsonite's pension fund commissioned Wells Fargo in 1971 to establish an index fund, allocating part of its assets to this arrangement. Research, client recognition, and the bank's execution formed an actual connection here. Wells Fargo's relatively limited existing active stock-picking client base also reduced conflicts between the new business and existing business claims (MacKenzie, 2006).

The function of index investing does not require every investor first to prove that all securities are correctly priced at all times. For investors who lack stock-picking advantages and value cost control and diversification, tracking a well-defined benchmark can be a conditional choice. The product solves the problem of market participation and controlling certain costs; it does not provide complete knowledge of every firm's future state.

As the index became an investment object and performance benchmark, it also changed the behavior of active management. MacKenzie (2006) discusses how some managers moved their portfolios closer to the index in order to reduce the career risk of falling significantly behind the benchmark. Here, not everyone was persuaded by the same theory; it is also possible that institutional incentives raised the cost of deviating from the benchmark. The market portfolio went from theoretical concept to operational reference, and then shaped holdings through performance evaluation.

In fiber language, observing portfolios becoming increasingly similar is compatible with at least two explanations: investors' judgments about returns and risks are converging, or different judgments still exist but assessment and career risk push managers toward similar positions. Reading portfolio similarity directly as belief identity misses the latter mechanism. The history of knowledge production must track both what participants believe and which judgments can change actual allocations.

§3.4 Professional Recognition and Actual Adoption Are Not the Same

Performance evaluation changed not only the interpretation of returns but also what performance could prove an institution to be professional. MacKenzie (2006) recounts a 1978 industry interview: Gary Bergstrom noted both that quantitative reports were entering offices and meetings in large numbers and that institutions actually using the new techniques to formulate strategies were still limited; he also observed that some managers hired quantitative staff to present to clients as a way of showing they had kept up with new methods. Here is a concrete transformation: knowledge tools originally used to test investment skill were also being used as signals to display investment skill.

This material supports not the claim that all quantitative institutions are mere appearance, but the claim that professional recognition and actual adoption are not the same. Clients seeing reports and team configurations can confirm that an institution has acquired certain resources, but cannot know from this alone whether those resources have entered position decisions. To distinguish the two states, one must examine the relationship among investment processes, position adjustments, and reports. However complete the report format, this evidence must still be obtained from processes and positions.

Index investing provided another certification path. As market benchmarks entered mandate evaluation, some managers reduced career risk by staying close to the index; trustees also feared being called imprudent after deviating from prevailing portfolio principles. In this environment, following modern portfolio theory could serve as a defense of prudent management. What is discussed here is the institutional expectations and behavior MacKenzie records, not a claim that law forced all funds to index.

Option theory used to answer gambling objections, quantitative tools used for client presentations, and portfolio principles used to defend prudent management show that the paths by which knowledge tools acquire qualifications are not uniform. Social recognition sometimes precedes full empirical testing, and sometimes accompanies partial functionality. These do not by themselves prove certification overreach, but they make the next question concrete: is what is being recognized a usable comparison tool, or a capability that has reliably completed a specific task; and who explains to the recipient the conditions separating the two?

§3.5 Paper Pricing Sheets Bring Formulas into the Trading Floor

The fact that option models can be solved on computers does not mean traders can complete calculations instantly in a busy trading pit. MacKenzie (2006) describes how the trading floor had rules restricting equipment use, and that entering data into a calculator and waiting could mean missing opportunities. The important vehicle that brought mathematics into the trading floor was, instead, paper.

Black generated theoretical prices on a computer outside the trading floor, reproduced them as paper tables, and offered them to subscribers. The tables listed option values under conditions of underlying price, strike price, expiration date, and so on, as well as price sensitivities to help adjust hedges. Traders could carry them and consult them quickly. The tables' size, information layout, and consultation speed made results that had required specialized computation into readings usable on the trading floor (MacKenzie, 2006).

This was not merely a change in how results were presented. The calculator was responsible for estimation and table generation; the service organization was responsible for delivery; the trader was responsible for identifying current conditions, looking up the table, and placing orders. Judgment, calculation, and execution began to be borne by different people and things. A trader who could not re-derive the equation might still use the table

accurately to complete a particular operation; but he needed to know what the table applied to, and someone needed to maintain the inputs and handle exceptions.

MacKenzie also discusses other pricing services, practitioners' resistance, and factors such as the model's public availability and the relative intelligibility of its parameters. The spread of the Black-Scholes model depended on a combination of theoretical content, accessibility, and practical work needs. Reducing it to the claim that any formula becomes true if only it is widely enough publicized deletes the most important selection conditions from this adoption history.

§3.6 Shared Scales and Distributed Judgment

When an institution simultaneously operates across multiple exchanges, multiple underlyings, and large numbers of options with different expiration dates, a single trader's experience is difficult to aggregate directly into an institutional position. Here the model acquires another use: enabling trades at different locations to be reported and compared on a shared scale.

MacKenzie (2006) records that institutional traders used tables to convert option prices into implied volatility, and used this reading for internal comparison and coordination. Implied volatility organized contracts with different underlying prices, strike prices, and maturities into a language usable in meetings and trading instructions. Shared readings reduced coordination costs, but comparison still depended on whether the assets and model conditions being compared were appropriate.

Implied volatility is not a directly observed future volatility. It is a parameter inferred from option prices given a pricing model and other inputs. It is therefore, first of all, a model parameter corresponding to current market prices, not a direct measurement of future actual volatility. Interpreting it as a prediction of future actual volatility further requires addressing risk compensation, supply and demand, model error, and term structure.

Beunza and Stark's ethnography of an arbitrage trading room further shows that opportunity identification and calculation occur within specific configurations of people and tools. Traders turn one security into a comparative scale for another; calculation is distributed across persons and instruments, rather than being completed in a single whole mind and then handed over to a machine (Beunza & Stark, 2004). This corroborates the history of paper pricing sheets and shared readings: models redistribute the locations of judgment; they do not thereby eliminate judgment.

§3.7 Automated Quotation Places the Model into Public Readings

In 1986, the Chicago Board Options Exchange introduced the automated quotation system Autoquote. MacKenzie (2006) records that its early use was related to the fact that certain stock quotations might not directly represent executable prices: the system input prices of relatively liquid options, computed synthetic underlying prices compatible with them, and used these to generate theoretical quotations for other options. At this point, the model was no longer merely a computational tool in individual hands, but part of the public quotation process.

This arrangement had practical uses: when direct quotations were insufficient, it organized quotations through relationships among tradable contracts. It also changed the sources of readings. A price seemingly about the underlying asset might already be the result of other option readings computed through the model. Evaluating the system requires preserving this chain of provenance. Multiple quotations agreeing with one another may

indicate that constraints were executed, but it does not necessarily mean that independent observations have increased.

From risk evaluation to index products, from paper tables to automated quotation, the process by which tools acquire qualifications for use always involves carriers and procedures. Procuring reports, displaying professional teams, forming shared readings, and actually changing positions cannot be conflated into a single kind of adoption. The next section asks: after these tools entered the market, to what extent can market outcomes still independently evaluate them?

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§4 How Market Evidence Changes after Adoption

§4.1 Models Provide Both Prices and Action Instructions

In 1973, Black and Scholes published their option pricing research, and the Chicago Board Options Exchange began trading standardized stock options in the same year. The near-simultaneous appearance of the formula and the new market provides a concrete scenario for observing how knowledge tools enter practice (Black & Scholes, 1973; MacKenzie, 2006).

A key feature of option theory is the use of dynamically adjustable positions in the underlying asset and financing to replicate derivative payoffs. When replication and the corresponding market conditions hold, the derivative price and the replication cost are constrained by arbitrage relations. This means pricing does not require first independently estimating the underlying asset's expected growth rate; but volatility estimation, transaction costs, financing conditions, and whether replication can be completed as assumed still require judgment.

The researchers thus provided two connected tools: one giving conditional theoretical prices, the other specifying how to adjust hedge positions. Pricing and action are directly linked here. Investors receive not merely a description of the world, but potentially a set of instructions capable of changing their own positions and trading needs.

Option theory also entered the process by which the new market sought recognition. MacKenzie's interviews show that Black-Scholes research helped the exchange explain option trading through hedging and efficient pricing, weakening the stigma associated with gambling. This is participants' explanation of the theory's institutional role, not proof that regulators had verified all risks. At the same time that theory helped secure legitimacy, its fit with actual prices was only approximate (MacKenzie, 2006).

§4.2 Use May Bring Price Relations Closer to the Model

After pricing sheets entered trading, participants could use the model to compare different options on the same underlying. MacKenzie (2006) discusses spread trading in particular: buying options that are relatively cheap according to the model and selling those that are relatively expensive, while arranging positions according to the relevant sensitivities. Compared with directly buying and selling the underlying stock for hedging, this approach avoids some transaction costs and reduces sensitivity to a single volatility estimate.

This strategy created a special connection between theory and empirical testing. The model requires that options on the same underlying with the same expiration but different strike prices have consistent implied volatility. The relative price differences that traders seek and exploit are precisely the deviations that researchers later

examine when testing the model. If trading activity narrows these differences, model use may help form price arrangements that better fit the model (MacKenzie, 2006).

This test has genuine discriminating power: the same parameter must also explain different contracts, so it is not a matter of fitting each price individually and declaring success. But it tests cross-contract price relations in specific samples; it cannot directly substitute for testing future volatility or replication capability under stress. If participants' trading did help form this relation, the new knowledge should also include which trading and institutional conditions maintain it.

MacKenzie did not treat the change in fit before and after as sufficient causal proof. He pointed out that other market changes must also be ruled out and specific behaviors traced; in the case of option spread trading, he preserved the corresponding causal interpretation as a judgment with grounds but not beyond doubt (MacKenzie, 2006). Market conditions were also changing: improvements in stock lending, commissions, price transmission, and execution methods made establishing and adjusting replication portfolios more convenient, and not all of these were caused directly by the model.

Fiber analysis thus separates "price cross-section compatibility" from "execution condition reliability": improvement in the former does not automatically rule out different failure states in the latter. Improved fit through use is a real possibility, but improved fit cannot, independently of model use, prove the model correct.

§4.3 Reverse Calculation Can Organize Judgment, but Cannot Generate Independent Evidence from Nothing

Forward portfolio analysis derives positions from expected returns, risk relations, and constraints. Reverse analysis asks: given an optimization model, risk estimates, and constraints, what expected returns would make the observed portfolio a reasonable choice? This is a conditional reverse computation. It can expose implicit assumptions and provide a comparative starting point for expressing new judgments.

Reverse calculation does not directly read investors' minds. If risk preferences, liabilities, financing constraints, or trading purposes differ, the same position can be compatible with different beliefs. Conversely, changing one of these conditions can yield a different set of implicit expectations. Labeling the result "market-implied" does not solve this identification problem.

Implied volatility is a similar case. Using an option price to back out a parameter and then plugging the same parameter back into the same formula to recover the original price only shows that reverse and forward computation are compatible. Declaring the model independently validated on this basis reuses the same piece of evidence. However, only some market calibration falls into this circularity.

Unlike back-calculation from a single price, cross-contract testing, holdout samples, and subsequent hedging errors can provide new constraints; these tests cannot be equated with re-substituting the same price, nor can all parameter adjustment be labeled closed-loop pseudo-calibration—the self-confirmation of validating a model against its own output without introducing external checks.

This chapter connects the financial industry's parameter calibration with calibration in the epistemological sense, but does not equate the two. The former usually means adjusting parameters to fit specified market data; the latter further asks what probative force that data has for the reliability being claimed. Market prices can be external inputs to a computation while still being influenced by the trading activity of widespread adoption of

the same model; the computation did not generate the price, but that does not mean the price is independent of the model's market use.

§4.4 Predicting Prices Is Not the Same as Revealing Information Needed for Action

Prices can also aspire to another status: guides to action. Managers refer to prices when deciding on investment, boards refer to prices when deciding on personnel changes, and regulators refer to prices when deciding on intervention. Bond et al.'s review points out that securities on secondary markets change hands among investors; capital does not flow to firms; for prices to affect the real economy, they must pass through real decision-makers (Bond et al., 2012).

This channel genuinely exists. Managers know their own companies best, but the success of a merger also depends on the target company's independent value, the synergies the merger can bring, and where the industry is heading. External conditions may include information that management does not yet know, and prices aggregating dispersed judgments may provide incremental material. Prices are therefore not only materials for estimating the future, but also inputs to action.

The effects of this channel depend on information, authority to act, and incentive conditions. Bond et al. distinguish forecasting price efficiency (FPE) from revelatory price efficiency (RPE): the former concerns whether prices can accurately predict future value, the latter concerns whether prices reveal the information necessary for decision-makers to take value-maximizing actions (Bond et al., 2012).

Imagine a company whose operations are deteriorating, whose board has the power to replace management, and whose market knows the board will act. If effective intervention can raise the post-event value of the deteriorated state to match that of the healthy state, the two situations may yield the same price for the same company: operations are acceptable and no intervention is needed; operations are deteriorating but rescue expectations are already priced in. The price predicts the company's value correctly in both situations, but it cannot answer which situation currently obtains or whether action is needed.

In fiber language: the observer is a board preparing to intervene, the reading is the price, and the same reading is compatible with two states requiring different handling. The way to shrink this fiber lies outside the price: looking at the same price again yields the same price; what is needed are orders, receivables, asset quality, and performance records. Forecasting price efficiency and revelatory price efficiency must be separated; those who need to act must still obtain independent information that can distinguish the relevant states.

Speculators knowing that decision-makers watch prices can itself change trading incentives, thereby changing what prices can reveal. Here there is no need to assume that participants mistakenly believe in prices, nor to judge every kind of feedback as pathological; what needs to be studied is how, after prices are used, information content and conditions for action concretely change.

§4.5 Publication and Adoption May Shrink Original Return Opportunities

The relationship between market outcomes and model use is not limited to improved fit. McLean and Pontiff's 2016 study of 97 return-predicting variables in the Journal of Finance compares three periods: the original sample, out-of-sample but pre-publication, and post-publication. Portfolio returns were 26% lower out-of-sample and 58% lower post-publication; the former is an upper-bound estimate of data-mining effects, and the 32%

difference between the two was estimated as the return decline associated with informed trading following publication (McLean & Pontiff, 2016).

This research supports a reverse proposition: investors may learn to identify mispricing from academic publications, and publication and adoption may thereby shrink original return opportunities. Declining performance after publication does not automatically mean the original research was falsified; it may also mean that knowledge dissemination changed the original opportunity. The rising correlation of predictor portfolios with other published predictor portfolios after publication also provides a further entry point for observing collective adoption.

This explanation cannot become a blanket exemption for any failure. Out-of-sample decay, post-publication decay, and other changes in market conditions must be handled separately; data mining, rational pricing, and the correction of mispricing each require their own evidence. More importantly, evaluating the original research cannot ask only "does it still make money later?" but must also ask whether the later return opportunities have already changed because of the research's dissemination.

This case contrasts with Chapter 23's evaluation of publication: publication can recognize and disseminate knowledge while simultaneously changing the opportunities that knowledge describes. It also differs from Chapter 18's skill competition: the core here is not that managers' improved skills generally shrink industry profits, but that public publication itself enters the formation process of subsequent evidence.

§4.6 Shared Verification Can Also Lose Independence

Beunza and Stark's three-year observation of a derivatives trading room shows that traders use models to convert prices into estimates of competitors' judgments, in order to check their own models. They call this practice reflexive modeling: prices are no longer merely inputs to one's own valuation, but also windows into how others understand (Beunza & Stark, 2012).

This arrangement can increase distributed cognition. Traders' models, information, and positions differ; market prices and others' behavior can provide constraints one has not yet seen. Knowledge does not exist only inside a single expert, but forms among people, models, and quotations.

But the same arrangement can also produce cognitive interdependence. When enough traders ignore the same key issue, their similar positions can provide each other with false reassurance: others seem to have reached the same conclusion, and a shared blind spot is mistaken for independent confirmation. A cautious reflexive process can thereby turn into overconfidence.

This complements the distinction from reverse calculation. Re-substituting the same price is a computational failure to add new evidence; shared blind spots involve new behaviors and quotations, but no new independent constraints. Judging whether market verification is effective requires not only counting how many people, models, and quotations there are, but also checking whether they bring different data, questions, and opportunities for falsification.

§4.7 Information Production Itself Also Requires Incentives and Procedures

Grossman and Stiglitz (1980) discuss market efficiency when information is costly: if prices already fully reflect the advantages obtainable from acquiring information, the incentive to bear information costs is affected. This

tension does not mean prices contain no information, but it requires studying who produces information, how producers are compensated, and how much information can enter public prices.

This question reconnects the preceding sections of this chapter. Databases require institutional investment, analysts require access and judgment, quotations require carriers and coordination, and public research requires dissemination. If one checks only whether prices contain information without checking the conditions of information production, one misses why knowledge is produced, why it is not produced, and which information has not entered prices.

After adoption, changes in market evidence run in at least four directions: use brings price relations closer to the model; publication and adoption shrink original return opportunities; shared blind spots weaken independent verification; and information production incentives change available materials. These are not synonymous repetitions of "model success causing failure." Before using market outcomes to certify a tool, one must explain how those outcomes were formed by the tool's use, and what independent verification remains.

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§5 Portfolio Insurance: Shared Execution Conditions, Risk Knowledge, and Layered Correction

§5.1 Different Performance States behind the Same Protection Floor

Portfolio insurance can advance the above questions to concrete fiber discrimination. Portfolio insurance is a strategy of adjusting asset positions to maintain a certain value floor; it can synthesize payoffs similar to a put option, but this does not make it equivalent to having purchased an option for which a counterparty bears payment obligations. MacKenzie (2006) records that the designers knew gap-downs could make replication too late to complete, and that they added checks and protective arrangements for specific gap-down scenarios for some clients.

First fix the observer and the task: a fund's trustee, when accepting a scheme, must judge what the scheme relies on to maintain the floor and what residual risks it must bear under stress. To compare mechanisms, suppose all schemes report the same target floor; this is not to assert that the historical materials contain several contracts with identical amounts, but to construct a contrast from existing mechanisms. If all arrangements are mapped onto a single floor amount, the different states in Table 2 cannot be distinguished from that amount.

Table 2. Protection Arrangements Compatible with the Same Target Floor

Protection arrangement	Mechanism relied on to maintain the floor	What the trustee must further check
Holding a protective put option matched to the underlying and maturity	Corresponding contractual rights and payment obligations	Whether the position exists, payment conditions, counterparty and clearing arrangements
Dynamically trading to replicate the protective payoff	Ability to adjust stock, cash, or futures positions as prices change	Transaction costs, financing, gap-downs, and execution ability during collective selling

Protection arrangement	Mechanism relied on to maintain the floor	What the trustee must further check
Dynamic replication with additional gap-down checks	Performance still depends on dynamic trading; checks identify applicable boundaries	Which gap-down, fill, and financing scenarios the checks cover, and whether results trigger follow-up action
Dynamic replication with cash exit or funding buffer rules	Changing positions or increasing available funds under agreed conditions	Trigger rules, execution ability, funding sources, and opportunity costs after exit

No arrangement in Table 2 is unconditionally optimal; directly purchasing options also has counterparty and clearing arrangements that need checking. What the table distinguishes is responsibility and execution structure. If contract verification finds no corresponding option position, the protection can no longer be explained as provided by that option's payment obligation; if trading stress checks find replication cannot be completed, the applicable scope of dynamic protection must be re-evaluated. Only when new evidence rules out specific explanations does fiber discrimination complete a step of actual work.

Knight's distinction between measurable and unmeasurable uncertainty supplies an older conceptual coordinate for the arrangement of Table 2 (Knight, 1921). Measurable uncertainty—for which Knight reserved the term "risk"—can be transferred through grouping and insurance: a larger contingent loss is converted into a smaller fixed payment to a counterparty that underwrites groups of cases. By this coordinate, the top row of Table 2 and the rows below it stand on opposite sides: holding a protective put is contractual transfer of risk, with the counterparty bearing the payment obligation; dynamic replication carries an unmeasurable residual, whose performance depends on the market's continued ability to absorb adjustments during declines—a condition that changes with the scale of adoption and cannot be priced or transferred by grouping into classes. The word "insurance" in portfolio insurance holds only in Knight's former sense.

Furthermore, checking and intervening cannot be conflated. Verifying a contract can make the trustee clearer about existing arrangements, but it does not automatically change them; adding funding buffers, adjusting positions, or changing trading limits can change the arrangements themselves. The former is discerning states originally merged under the same floor; the latter is changing the conditions of future performance. Merely computing the floor figure more precisely can substitute for neither the first nor the second.

This distinction also shows how to judge the pseudo-small fiber. If the trustee already knows the floor depends on dynamic execution and accepts the residual risks in light of his own obligations, he cannot be judged as deceived merely because risks exist. Only if he treats a goal that depends on trading as a certain payment requiring no further checking of performance conditions does a gap form between perceived certainty and actual conditions. Existing materials can support revealing this risk and several failure mechanisms; they cannot attribute the same psychological state to every historical client.

§5.2 How Local Protection Changes Shared Execution Conditions

Once floor arrangements are adopted, another problem emerges with scale. For a small dynamic strategy, the market may suffice to absorb adjustments; for many institutions simultaneously executing similar strategies, that

absorption capacity cannot be maintained as an unchecked premise. The more portfolio insurance is adopted as a protective instrument, the more same-direction selling may be triggered in future downturns. If selling affects prices, new declines trigger additional selling, and protective instructions participate in changing the very market on which protection depends. This change in shared execution conditions falls within the range of Knight's unmeasurable uncertainty—situations too unique to be grouped or assigned an outcome distribution—and cannot be eliminated by consolidating more cases of the same kind (Knight, 1921).

This is not a result conjured from the form of feedback alone. MacKenzie (2006) records that Bruce Jacobs had already raised concerns in a 1983 internal memo that widespread adoption could snowball price movements; the designers later also attended to the market impact of their own trading scale. What can be established here is that the mechanism and the scale problem had already been raised at the time; it cannot be directly written as a definitive attribution of the 1987 decline.

In the fiber framework, the price and position records originally used to evaluate local execution may not suffice to discern feasible states under collective adjustment. Even if the model's equations have not changed, the scale of users, other institutions' positions, and counterparties' behavior can change the relationship between outputs and consequences. This is the specific content that dynamic corrosion needs to test in this case: it is not that readings necessarily go bad once used, but that use may change the conditions that originally supported their reliability.

Pathological recursion therefore does not require anyone to have explicitly refused testing. Even if each firm computes its own exposure, if those computations collectively depend on others being able to absorb the trades, and if competition pushes similar strategies to keep expanding, local defense can still increase systemic fragility. Success being used to suppress dissent is one mechanism; dispersed decision-making being unable to maintain shared conditions is another; the two can overlap or occur separately.

Diagnosis must be checked against the specific task. Spread trading improving relative pricing does not guarantee that large positions can be exited under stress; protecting one client cannot by itself prove that risk has not been transmitted to other market participants. Conversely, the system being under stress does not cancel the protection some clients actually received. Local gains and collective consequences must be stated separately before one can judge what feedback improved and what it damaged.

§5.3 Why Recognized Risk Failed to Change Adoption

Participants in portfolio insurance did not first hear about limitations only at the crisis. MacKenzie (2006) records that the designers had worried that taking on a large pension client would bring excessive market impact, but also believed that refusing would send the client to a competitor. One company reducing its business cannot guarantee that other companies' similar positions will correspondingly decrease; the incentive to limit one's own business and the need to reduce collective risk are two different things.

The dissemination of risk judgments was also jointly influenced by interests and evidentiary conditions. Rubinstein recalled in a later interview that after the 1986 market decline, he suspected that portfolio insurance's futures selling pressure played an important role, and he expressed his views to the SEC but did not publish his concerns; he cited both uncertainty and reluctance to harm the business as reasons. The regulator's judgment of the decline's causes at the time differed from his (MacKenzie, 2006). This material cannot prove that his causal judgment was correct, nor can it be written as though he never reported the risk; it shows how undetermined

risk simultaneously faces demands for justification and commercial costs.

Four situations must be distinguished here: risk not recognized, risk recognized but lacking decision authority, decision authority present but constrained by competition, and individual measures unable to maintain shared execution conditions. Merely adding a risk disclosure may improve the first situation, but it does not necessarily change the other three. Calling all of them "participants mistakenly believing in the model" omits the organizational and institutional conditions needed to translate knowledge into action.

Continuing to compute one's own positions and having the authority to check and limit collective selling volume are different capabilities. Whether risk knowledge enters business decisions depends on whether it can affect mandate scope, client acceptance conditions, and inter-institutional constraints, not merely on whether reports mention the risk.

This also makes the comparison between finance and academic evaluation more concrete. Both may reward performance produced according to established rules, but financial adoption can also change the next round's execution environment through positions. The gap in knowledge production can be either interpreting achievements too fully, or having objections already appear without being able to effectively change the arrangements that keep producing those achievements.

§5.4 Execution Divergence and Client Outcomes in the 1987 Crash

In October 1987, the shared execution conditions of portfolio insurance were directly tested. Execution, quotation synchronization, and financing constraints meant that the same protection goal no longer corresponded to a single operational instruction. MacKenzie (2006) discusses portfolio insurance selling, other stop-loss strategies, cross-market transmission, and infrastructure pressure, while explicitly preserving the difficulty of attributing exactly how much portfolio insurance amplified the decline.

Price differences themselves also cannot directly issue a single action instruction. Futures trading at a large discount to the spot index could mean futures were being depressed by mechanical selling, or it could mean some stocks had failed to trade in time and the index was using stale prices. MacKenzie records that Leland O'Brien Rubinstein Associates, the firm founded by portfolio insurance's designers, slowed its selling amid concerns about market impact and price differences, while Wells Fargo Investment Advisors continued selling futures aggressively. The same market anomaly entered different judgments and executions; one cannot evaluate solely by who was more faithful to the formula.

This brings the discrimination of protection floors down to the event process: the materials needing verification shift to whether orders could be filled, which time points' prices composed the index, whether financing was available, and how much unhedged risk would remain after slowing trades. Judging these conditions is not a low-precision substitute for computation; it is the work of deciding when computational outputs can still be executed.

Knight's account of profit and judgment supplies the oldest formulation of this predicament. In his view, profit arises precisely where probability calculation is impossible; an individual success cannot evaluate judgment, and only a succession of cases suffices to assess the probable value of a judgment—by which time the profit has already been transformed into a wage. He wrote:

> The receipt of profit in a particular case may be argued to be the result of superior judgment. But

> it is judgment of judgment, especially one's own judgment, and in an individual case there is no way of telling good judgment from good luck, and a succession of cases sufficient to evaluate the judgment or determine its probable value transforms the profit into a wage. (Knight, 1921)

Knight's classification in the same chapter runs on three levels: probability judgments rest either on completely homogeneous classification of instances (a priori probability) or on empirical frequency (statistical probability); and estimate—which Knight calls a third type of probability judgment—must be "radically distinguished" from both. He wrote:

> The liability of opinion or estimate to error must be radically distinguished from probability or chance of either type, for there is no possibility of forming in any way groups of instances of sufficient homogeneity to make possible a quantitative determination of true probability. Business decisions, for example, deal with situations which are far too unique, generally speaking, for any sort of statistical tabulation to have any value for guidance. The conception of an objectively measurable probability or chance is simply inapplicable. (Knight, 1921)

It is this third type of probability or uncertainty which has been neglected in economic theory; its income form is the final definition of profit—

> It is the margin of error in this most ultimate faculty of judging faculties whose exercise is the essence of responsible control, which constitutes the only true uncertainty in the workings of the competitive organization (as of any other organization). And it is uncertainty in this sense which explains profit in the proper use of the term, the sense toward which economic usage has been groping, that of a pure residual income, unimputable by the mechanism of competition to any agent concerned in its creation. (Knight, 1921)

The later fate of this distinction in economics is documented in histories of thought. Chen and Liu (2020) place Knight, alongside Keynes, Hayek, and Shackle, in the line of uncertainty thought descending from Smith and Menger—they acknowledge that uncertainty is the fundamental and essential problem of the economy, the principal motive force of economic evolution, and the source of its unquantifiability; opposed to it is the "probabilist" direction, which brings uncertainty into mathematical models through probability. The subsequent evolution their survey compresses into one sentence:

> The integration of subjective expected utility theory over rational expected utility and subjective probability completely ignored Knight's distinction between risk and uncertainty, leading risk and uncertainty in mainstream economics to converge toward identity, uncertainty being interpreted as a computable form of risk, the only difference being that risk is taken as objective while uncertainty is taken as subjective. (Chen & Liu, 2020; author's translation)

In other words, the boundary Knight drew—the non-groupability of estimates, the place of judgment—is precisely the line later crossed by probabilistic modeling; the whole discussion of judgment, models, and market feedback in this chapter takes place on the two sides of that crossed line.

The execution divergence of October 1987 sits precisely in the territory of the evaluation of judgment: that Leland O'Brien Rubinstein Associates slowed its selling while Wells Fargo Investment Advisors continued selling aggressively cannot, after the fact, be adjudged as judgment versus luck within a single event; and the succession of cases sufficient to evaluate judgment can be accumulated only at the cost of execution records

from different institutions under repeated stress. There is no shortcut to the evaluation of judgment, and this has nothing to do with the sophistication of the tools.

Existing attribution evidence is insufficient to assign the entire crash to a single formula. Other stop-loss strategies, widespread crash expectations, trading infrastructure, and international market linkages all make single attribution untenable. What this case more directly presents is the testing of execution conditions under stress: the scale of protection demand, trading direction, and market absorption capacity entered the process of whether the method could continue to fulfill its promises.

Local protection and systemic impact must also be distinguished. MacKenzie records that some clients still received considerable protection during the sharp decline, while others' portfolios fell below their stated floors; some portfolios that shifted to cash then faced the cost of missing the rebound. These different outcomes cannot be canceled by a single label of overall success or failure. Client losses, strategy execution, market feedback, and subsequent opportunity costs must be evaluated separately.

§5.5 Shared Language Can Record Anomalies

After 1987, a persistent implied volatility skew appeared in the U.S. equity index options market. Different strike prices no longer displayed the flat relationship required by the classical constant-volatility model. Traders continued to use implied volatility language, but used the skew to describe and arrange relative prices of different contracts; quotation systems also absorbed the new treatment (MacKenzie, 2006).

The continuation of a shared language does not mean the same assumptions went untested. MacKenzie records that market makers could set and adjust the skew, automated quotation systems combined it with corresponding computational methods, and traders discussed the skew's response to different scenarios. Here the implied volatility language was preserved, but the flat relationship required by the classical model was not. Model language can both organize trading and record deviations between model and prices.

Chapter 20's analysis of the naturalization of model presuppositions asks how shared language limits the questions that can be raised. This section need only preserve a more concrete distinction: shared language, fixed assumptions, and authority to judge are not the same object. The same language can accommodate revisions of parameters and methods, but which questions can enter decisions still depends on specific procedures. A more specific limitation occurs when institutions only allow discussion of how to restore quotation fit, while objections concerning execution, collective risk, and client obligations cannot change business decisions.

One must therefore check which layer a correction can reach. Changing the skew can improve current quotations and may also change trading and hedging; capital and margin adjustments directly change the positions and funding constraints one can bear. All of these have practical effects, but their objects differ. Whether the former kind of improvement simultaneously improves the latter two conditions requires further checking; it cannot be replaced by noting that the shared language is still in use or has already changed.

§5.6 How Risk Awareness Enters Shared Rules

MacKenzie (2006) also records that after the crisis, firms providing clearing services to traders changed their risk checks, assessing the consequences if traders encountered similar shocks again; the Options Clearing Corporation began using stable distributions to set margin requirements in 1991. Here is a crossover important

to the history of theory: the fat-tail methods that had challenged mean-variance computation entered another institutional task—deciding how much protection to reserve for extreme movements.

This is not contradictory. Comparing portfolios by variance and setting margins by tail quantiles require different mathematical quantities. A distribution lacking finite variance does not thereby lack meaningful quantiles. The succession of knowledge tools is therefore not a single-line replacement; different methods can coexist in the same market for different tasks.

For constraints to take effect, competition among institutions is also involved. MacKenzie quotes risk management participant Timothy F. Hinkes's explanation: in the past, if a clearing firm demanded excessively strict terms, traders could switch to firms offering looser conditions; more uniform capital and margin requirements made it less easy for individual firms to lose business when limiting risk. What changed here is not only the risk model, but also the conditions under which recognized risk can enter constraints.

This material contrasts with the designers' fear of client defection. In this competitive structure, merely increasing risk awareness does not guarantee that limits will take effect; more uniform requirements reduced the space for competing for business through looser conditions, making it easier for some risk judgments to become action. The new requirements also changed who had the capital to continue providing protection; gains in safety and the costs of entry and changes in organizational structure must all enter the evaluation.

Institutional adoption also does not mean risk has been eliminated. MacKenzie discusses the mechanisms by which more conservative tail specifications weaken procyclical margin changes, but one cannot attribute all subsequent stability to a single model merely because the crisis did not repeat in the same form. For this chapter, the clearer change is that judgments about jumps and tails did not remain only in papers, but entered the constraints faced by traders and their clearing firms.

§5.7 Repairing the Model and Repairing the Conditions of Use Are Not the Same Thing

Distribution research, price anomalies, and crisis experience each exposed different gaps. Replacing distributions can change tail estimates, adjusting parameters can improve quotations, checking execution can reveal trading constraints ignored by the model, and uniform risk requirements can reduce the space for institutions to evade constraints by switching business. These are not different names for a single universal calibration.

This chapter treats these changes as three interrelated kinds of work. The first is making data and computation more accurate, reducing detectable errors. The second is expanding the scope of the problem, bringing previously unrepresented execution, financing, and collective risks into judgment. The third is changing adoption conditions, so that new understanding can affect positions, scale, and responsibility, rather than appearing only in research reports.

Testing these three kinds of work against portfolio insurance's protection task makes the distinction clearer. Recomputing theoretical positions first checks whether the computation is faithful to the given assumptions; adding gap-down, fill, and financing scenarios tests the original scope of protection; writing the check results into scale limits and collective risk requirements further changes actual adoption. The three cannot substitute for one another, and they need not all be completed by the same institution.

At the same observation point and under the same task, new evidence ruling out a certain performance state can be said to shrink the compatible range; modifying contracts, funding, or market rules changes the object

being analyzed itself; the resulting risk reduction cannot be called a shrinking of the original fiber. The fiber framework here helps distinguish epistemic improvement from object change, so that different reforms can be evaluated by their actual effects.

Conversely, limiting positions may reduce some systemic risk without proving that the model predicts the future more accurately. The achievements of knowledge, execution, and governance must be stated separately.

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§6 Conclusion: What Can Still Serve as Evidence of Knowledge Growth

§6.1 Functionality and Pathology Cannot Be Divided by Era or Model Name

From portfolio selection to risk adjustment, from index products to option pricing sheets, and then to shared quotation and post-crisis risk rules, the changes in financial knowledge are not merely a rise in formula complexity. More important is how judgment was divided among different people, organizations, and devices, and how market outcomes subsequently acquired recognition. Knowledge tools both entered action and entered the standards by which actors were evaluated.

The same arrangement can simultaneously serve two functions. An index can offer some investors choices in diversification and cost, and can also push portfolios toward convergence through career assessment; a pricing sheet can make complex computation usable, and can also prevent recipients from seeing input conditions; market calibration can make quotation relations more consistent, and can also be over-interpreted as tail risk having been controlled. One cannot judge functionality or pathology by the tool's name alone.

The fiber framework requires opening up the differences behind these identical readings. Recursive diagnosis simultaneously checks two paths: first, whether local achievements have acquired recognition for capabilities that were never tested; second, whether, even after risks have been recognized, dispersed decision-making and competition are still weakening shared execution conditions. Neither subjective credulity alone nor post-hoc losses alone can determine a pathological slide. What must be traced is how adoption changed risk, and which calibrations and constraints failed to take effect accordingly.

This criterion does not presuppose that pathological recursion has already overwhelmed functional recursion in financial markets as a whole. To judge whether pathological recursion has overwhelmed functional recursion, one must first specify whether the comparison is trading volume, resource control, risk exposure, or social consequences. This chapter's cases support identifying several self-reinforcing and corrective mechanisms; they do not provide a net-effect total covering the entire market.

§6.2 Financial Verification Conditions and Their Limits

Financial markets are not without opportunities for verifying promises. Dividends, interest, repayments, maturity payments, and actual transactions can rule out certain previously compatible states. Whether a protection was executed, whether a payment was received—these usually have much clearer objects of verification than a vague narrative of market prosperity.

But actual outcomes must also correspond to prior claims. A disaster that did not occur proves nothing about probability judgments; a single loss does not prove all related models wrong. Value, risk, and strategy returns

each have their own time ranges and comparison conditions; one cannot use today's price to settle future solvency, nor use final solvency to conceal the consequences of being unable to finance or exit midway.

Verification must also preserve the original task. Portfolio insurance exiting to hold cash and maintain a certain floor is not the same achievement as a pension fund continuing to meet years of benefit payments. Whether the floor was lowered after the crisis, whether risk was re-assumed, and who made up the shortfall all require fresh judgment; they cannot be decided solely by the tool supplier's performance report. This is not to prejudge that the relevant funds ultimately failed to pay, but to distinguish the accounting endpoint of the strategy from the duration of fiduciary responsibility.

The validity of financial knowledge therefore is not maintained by finding a final clock that never errs. It depends on the continued existence of verification at different levels: computation accepting review, estimates accepting new data, strategies accepting actual execution, and institutions having to answer what risks are worth bearing, who can modify objectives, and who bears failure. Professional division of labor can reduce the cost of these tasks, but it cannot quietly replace the purposes of the mandate with the convenience of the model.

§6.3 Four Kinds of Outcomes Settled Separately

This chapter's materials require evaluating four kinds of outcomes separately.

The first is knowledge growth. New independent constraints appear, previously compatible explanations are ruled out, boundaries of applicability are clarified, or errors are corrected along chains of use. Data encoding and cross-checking, cross-contract testing, holdout samples, execution records, and effective counterevidence can all serve this function. Knowledge growth is not more materials, more complex models, or more refined language, but states that were previously indistinguishable becoming distinguishable, or original errors and boundaries genuinely changing subsequent judgments.

The second is organizational coordination. Shared scales, databases, and procedures reduce coordination costs, enabling different people and institutions to compare, quote, and execute. Paper tables and shared language can produce genuine coordination gains, but shared language itself does not guarantee shared truth, and organizational survival does not prove methodological validity.

The third is commercial and professional success. Tools are procured, adopted, profitable, or used to display professional qualifications. These results can change resources and actions, but they cannot by themselves prove knowledge growth. A theory being procured and actually entering positions must be separated, and client recognition and substantive testing must also be separated.

The fourth is risk and execution improvement. Contracts, financing, margins, scale limits, or shared rules change actual performance conditions. Modifying these arrangements changes the object being analyzed and the governance conditions; the resulting risk reduction cannot be retroactively read as the original fiber having shrunk. Only when these changes are simultaneously grounded in new evidence, effective counterevidence, or clarified boundaries of applicability can one also speak of knowledge growth in the corresponding parts.

The four kinds of outcomes can support one another, but they cannot prove one another. Recording them separately prevents writing local success as overall validity, and also prevents writing local failure as zero knowledge gain.

§6.4 What Independent Verification Must Be Preserved after Successful Use

This chapter's central feedback is: local success may help a tool gain wider use, and wider use may change the conditions on which that local success depended. Its occurrence does not require all participants to have mistakenly believed in the model, and it can still be corrected. Shared language can record anomalies, and shared rules can make dispersed risk judgments acquire binding force; whether either works must still be tested separately.

This is the connection between financial clocks and the Clock Room: markets need readings that can be transmitted, compared, and executed, but readings cannot by themselves vouch for the entire process that produced them. The fiber framework discerns performance states not yet distinguished by the same reading; recursive analysis traces how use changes those states. For those who live and operate through financial arrangements, the final verification must still return to the obligations undertaken and the actions that can be sustained, and cannot stop at the clock showing well.

If this chapter selected only losses and crises, it would ignore the history of tools improving decisions; if it selected only adoption and profits, it would ignore the failure conditions that the same achievements have not yet ruled out. Markowitz's account of approximation boundaries, the production processes of data and analysts, the actual coordination gains of option trading, and the post-crisis changes in risk rules should all be preserved as materials capable of constraining the diagnosis, rather than being canceled at the end by an overall judgment of degradation.

This analysis also does not provide investment advice. It judges how large a knowledge claim the evidence can support; it does not decide which asset is worth buying at a particular moment. Maintaining vigilance against a self-certification mechanism and knowing the direction of the next price movement are entirely different capabilities.

Part IV's subsequent studies of platforms and AI may similarly change behavior through use and then evaluate themselves using the changed behavior. What financial history offers is not a conclusion that assigns them to pathology in advance, but four comparative questions: what is the knowledge product and its task here; by what does it acquire qualifications for use; how does use change subsequent evidence and execution conditions; and who can still independently verify, modify, or stop it. A tool's capacity to organize activity and its capacity to correctly understand that activity and its consequences must be tested separately. Preserving this distinction is what prevents the recognition of success from becoming a reason to stop judging.

References

Chen, X., & Liu, S. (2020). The Evolution of Economics' Understanding of Uncertainty [In Chinese]. *Caimao Jingji [Finance & Trade Economics]*, 2020(7), 3–13. https://doi.org/10.19477/j.cnki.11-1077/f.2020.07.001

Beunza, D., & Stark, D. (2004). Tools of the Trade: The Socio-Technology of Arbitrage in a Wall Street Trading Room. *Industrial and Corporate Change*, 13(2). https://doi.org/10.1093/icc/dth015

Beunza, D., & Stark, D. (2012). From Dissonance to Resonance: Cognitive Interdependence in Quantitative Finance. *Economy and Society*, 41(3). https://doi.org/10.1080/03085147.2011.638155

Black, F., & Scholes, M. (1973). The Pricing of Options and Corporate Liabilities. *Journal of Political Economy*, 81(3). https://doi.org/10.1086/260062

Bond, P., Edmans, A., & Goldstein, I. (2012). The Real Effects of Financial Markets. *Annual Review of Financial Economics*, 4. https://doi.org/10.1146/annurev-financial-110311-101826

Brown, L. D., et al. (2015). Inside the "Black Box" of Sell-Side Financial Analysts. *Journal of Accounting Research*, 53(1). https://doi.org/10.1111/1475-679X.12067

Fisher, L., & Lorie, J. H. (1964). Rates of Return on Investments in Common Stocks. *The Journal of Business*, 37(1). https://www.jstor.org/stable/2351197

Grossman, S. J., & Stiglitz, J. E. (1980). On the Impossibility of Informationally Efficient Markets. *American Economic Review*, 70(3). https://www.jstor.org/stable/1805228

Knight, F. H. (1921). *Risk, Uncertainty and Profit*. Boston: Houghton Mifflin.

MacKenzie, D. (2006). *An Engine, Not a Camera: How Financial Models Shape Markets*. Cambridge, MA: MIT Press.

Markowitz, H. M. (1952). Portfolio Selection. *Journal of Finance*, 7(1). https://doi.org/10.1111/j.1540-6261.1952.tb01525.x

Markowitz, H. M. (1959). *Portfolio Selection: Efficient Diversification of Investments*. New York: Wiley.

Markowitz, H. M. (1991). Foundations of Portfolio Theory. *Journal of Finance*, 46(2). https://doi.org/10.1111/j.1540-6261.1991.tb02669.x

Markowitz, H. M. (2002). Efficient Portfolios, Sparse Matrices, and Entities: A Retrospective. *Operations Research*, 50(1). https://doi.org/10.1287/opre.50.1.154.17774

Markowitz, H. M. (2020). *Risk-Return Analysis, Volume 3: The Theory and Practice of Rational Investing*. New York: McGraw-Hill.

McLean, R. D., & Pontiff, J. (2016). Does Academic Research Destroy Stock Return Predictability? *Journal of Finance*, 71(1). https://doi.org/10.1111/jofi.12365

Sharpe, W. F. (1964). Capital Asset Prices: A Theory of Market Equilibrium under Conditions of Risk. *Journal of Finance*, 19(3). https://doi.org/10.1111/j.1540-6261.1964.tb02865.x